Enhanced Expense Tracker - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

The Enhanced Expense Tracker is a financial management system built on a strict 3-layer architecture. The project utilizes SQLite for data persistence and mandates a Test-Driven Development (TDD) workflow. The current documentation serves as a quality validation gate, ensuring functional requirements remain technology-agnostic while adhering to a predefined project Constitution.

1.2 Maturity Assessment

The project is currently in a state of REFINEMENT. While the quality checklist indicates that functional requirements are verified and technology-agnostic, there is a high-severity structural gap regarding the detailed Scope and Out-of-Scope definitions, which are external to this validation document and must be cross-referenced from the primary specification.

1.3 Technical Stack

- SQLite

1.4 Architectural Constraints

- 3-layer architecture pattern.
- TDD (Test Driven Development) mandatory process.
- Strict separation of functional requirements from implementation details.
- Requirement testability and measurable success criteria.
- Explicit identification of edge cases and bounded scope.

1.5 Critical Dependencies

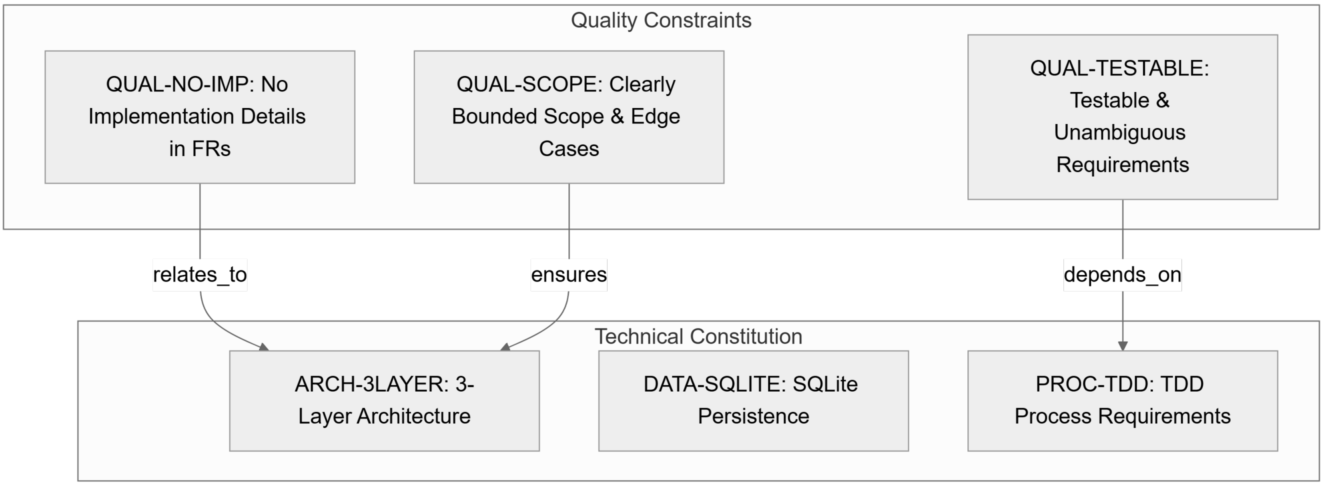
- SQLite schema for data persistence.
- TDD workflow depends on the existence of testable and unambiguous requirements.
- Alignment with the project Constitution for architectural integrity.

2. Architecture Workflows & Visual Diagrams

2.1 Specification Quality Validation Workflow

[Schéma non compilé: Diagramme #1]

2.2 Quality Constraints & Technical Requirements Traceability



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Type	Description	Source Section	Attributes
QUAL-NO-IMP	Constraint	Interdiction d'inclure des détails d'implémentation (langages, frameworks, APIs) dans les exigences fonctionnelles ou les critères de succès.	Content Quality	scope: Functional Requirements / Success Criteria
QUAL-TESTABLE	Constraint	Les exigences doivent être testables, non ambiguës et les critères de succès doivent être mesurables.	Requirement Completeness	status: verified

QUAL-SCOPE	Constraint	Le périmètre (scope) doit être clairement délimité et les cas limites (edge cases) identifiés.	Requirement Completeness	-
ARCH-3LAYER	Non-Functional Req	Adoption d'une architecture à 3 couches conformément à la Constitution du projet.	Notes	category: Architecture
DATA-SQLITE	Non-Functional Req	Utilisation d'un schéma SQLite pour la persistance des données.	Notes	category: Storage
PROC-TDD	Non-Functional Req	Application des exigences de Test Driven Development (TDD).	Notes	category: Process

3.2 Security Rules

No specific security rules defined in the provided source data.

3.3 Data Models

• **Persistence Layer**: SQLite schema (as mandated by `DATA-SQLITE`).

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Scope & Out-of-Scope	HIGH	Ce document est une checklist ; le périmètre détaillé doit être extrait du document spec.md mentionné.
Open Questions & Uncertainties	LOW	La checklist indique que tous les marqueurs [NEEDS CLARIFICATION] ont été supprimés.

4.2 Remediation & Workflow

The project must cross-reference the `spec.md` file to finalize the Scope and Out-of-Scope definitions. Once the structural gaps are filled, the specification will be fully aligned with the project Constitution and ready for the planning phase.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
Feature	BUSINESS_DOMAIN	Feature Readiness	A distinct piece of functionality that must satisfy measurable outcomes and a set of clear acceptance criteria.

TDD	TECHNICAL_STACK	PROC-TDD	A software development process where automated verification scripts are authored prior to the actual logic implementation.